

Investigating the Oscillation of a Pendulum

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Abstract—Finding the degree to which small angle approximations can be used for a pendulum and calculating the coefficients of higher order terms in the series expansion for period from experimental data. Using video processing tools, the pendulum's time period was plotted against its amplitude. The first term coefficient was $\alpha = 0.070 \pm 0.007$ agreeing with the theoretical coefficient 0.063. Additional analysis was done using a discrete fast Fourier transform, it was found that the pendulum was slightly longer than initially measured which led to a systematic error in the period-amplitude graph.

I. INTRODUCTION

PENDULUMS were first used as time keepers with the invention of the pendulum clock in 1656 by Christiaan Huygens. They increased the accuracy of time keeping by 60 times over previous mechanical methods. [1] Newton deduced that the earth was oblate after it was noted that pendulum clocks ran 5 minutes slower a day in French Guiana compared to Paris. Demonstrating their sensitivity to gravity and contributing to their subsequent use as gravimeters.

The motion of an ideal pendulum was not as simple as it first appeared. Simple harmonic motion would imply that time period has no dependence on amplitude, however for large angular displacements there was a noticeable difference compared to the predicted constant.[2]

The aim of this experiment was to validate the equations of motion for higher order terms of a series expansion for θ_0 when the angle becomes too large for small angle approximations to be used.

II. THEORY

Making use of classical mechanics, we can derive from Newton's 2nd law the equation of motion for a pendulum with bob mass m as:

$$m \frac{d^2\theta}{dt^2} = -m \frac{g}{l} \sin(\theta) \quad (1)$$

Where m is the mass of the bob, g is the gravitational field strength, l is the length of the pendulum and θ is the angular displacement from the stable equilibrium position.

If we assume that θ_0 is sufficiently small we can approximate $\sin(\theta) \simeq \theta$. Leading to the equation of motion:

$$\theta(t) = \theta_0 \cos(\sqrt{\frac{g}{l}} t) \quad (2)$$

Where θ_0 is the initial angular displacement (amplitude of oscillation).

However, if we want to solve the equation of motion analytically we use conservation of energy to derive (3).

$$\frac{d\theta}{dt} = \sqrt{\frac{2g(\cos(\theta) - \cos(\theta_0))}{l}} \quad (3)$$

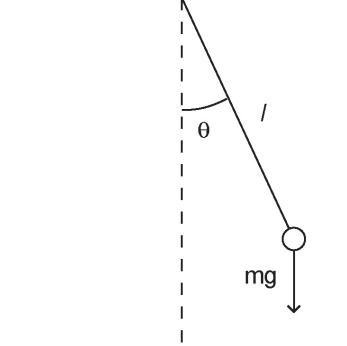


Fig. 1. Skeleton model of the pendulum, figure reproduced from [3]

Making a couple of substitutions we arrive at an expression of T in terms of a complete elliptic integral of the first kind.

$$k = \sin\left(\frac{\theta}{2}\right) \quad (4)$$

$$\sin(u) = \frac{\sin(\frac{\theta}{2})}{\sin(\frac{\theta_0}{2})} \quad (5)$$

$$T = 4\sqrt{\frac{l}{g}} \int_0^{\frac{\pi}{2}} \frac{du}{\sqrt{1 - k^2 \sin^2(u)}} \quad (6)$$

Using the Legendre polynomial solution for the elliptic integral and the Taylor expansion of $\sin(\theta)$ we arrive at the explicit equation:

$$T = 2\pi\sqrt{\frac{l}{g}} \left(1 + \frac{1}{16}\theta_0^2 + \frac{11}{3072}\theta_0^4 + \frac{173}{737280}\theta_0^6 + \dots\right) \quad (7)$$

We will be verifying these coefficients by recording the oscillations of a pendulum at multiple initial angles and optimising a fit to match the data.

The equation that will be optimised is:

$$T = 2\pi\sqrt{\frac{l}{g}} (1 + \alpha\theta_0^2 + \beta\theta_0^4) \quad (8)$$

As the pendulum is symmetric across its line of equilibrium, the time period should be the same for positive and negative θ which implies that no odd power terms exist.

III. METHODS

To setup the equipment in a way that minimised error we setup a camera on a clamp looking orthogonally down at a table covered by a matte black sheet. Making sure the scene is properly illuminated we then attach a pendulum to a clamp and set it directly below the camera. Where the plane of oscillation

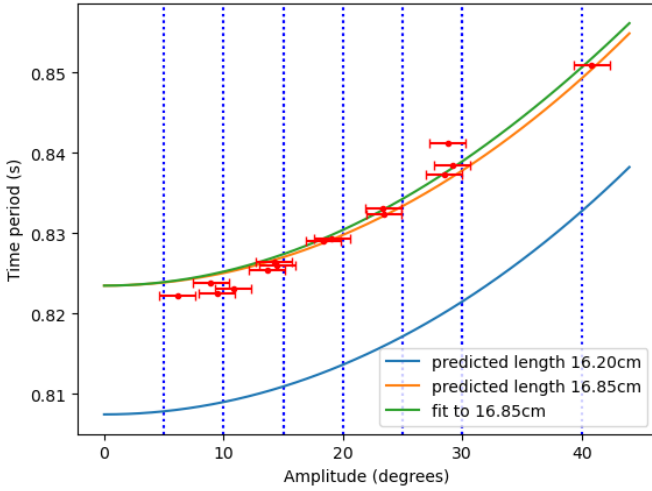


Fig. 2. A plot of fitted time period against amplitude of oscillation. It shows a relationship consistent with the theoretical model. Albeit with a systematic error. The vertical blue lines were the target angular displacements, however due to human and random sources of error the actual angular displacements are shown with the red dots. Each with a couple of degrees of error derived from uncertainty in the proportionality constant.

is in the x axis of the camera's view. We measured the length of the pendulum between its pivot and mass with a ruler.

Then we calibrated the video feed by finding a proportionality constant between $\sin$ of the angular displacement and the displacement in pixel coordinates on the sensor.

$$\theta = \sin^{-1}(kA) \quad (9)$$

Where k is the proportionality constant, A is amplitude in pixels, and θ is the amplitude in degrees.

In order to extract information from the camera we took videos of the pendulum oscillating starting from different angles. Then using the python library *cv2* we converted the video to black and white, flooring any pixels not reaching a threshold brightness to black. Thus only the pendulum is seen in the video as a white region. Extracting the coordinates of the white pixels and averaging them for each frame we generated an array of coordinates for the pendulum.

To keep the analysis simple we only focussed on the x coordinate array. It was seen that the x and y coordinates oscillated in phase and so only the x axis was needed to calculate time period.

Performing this time period analysis for increasing angles, we collected a data set of time periods against amplitudes. Then plotting that graph we compared it to our theoretical prediction produced from 7. And optimised the α and β coefficients from 8. As seen in 2.

Further more we investigated behaviour of the pendulum in Fourier space. Making use of *Numpy's* discrete fast Fourier transform and using *Scipy* to find the peaks in the graph.

IV. RESULTS & DISCUSSION

Firstly we measured the constants of the experiment. We found the length of the pendulum between the pivot and mass as

$$l = 0.162 \pm 0.002m \quad (10)$$

Using the library *scipy.optimize* we fitted a sin curve to the data 3, and used its parameters and covariance matrix to extract the time period and its uncertainty for each video. Only the first couple of oscillations were included in the fit to limit the effects of damping from air resistance on the system.

Then plotting a graph of Time period against Amplitude 2 it is clear that there is a relationship between the two. However whilst the data matches the general shape predicted by the theory, it is systematically shifted up by about 0.012s. Investigating the cause of this shift it was deduced that the length of the pendulum could have been measured incorrectly. Using *scipy.optimize* to find the length of the pendulum given the assumption that the theory is correct gave a length.

$$l = (0.169 \pm 0.001)m \quad (11)$$

This being 0.65cm larger than our measured value.

Additionally, fitting equation 8 with the length as 0.162m gave a very clearly bad graph which started in the same place as the blue line, then rapidly increased bisecting the data points and not matching their shape.

Adjusting the assumed length of the pendulum to 0.1685m we generated the orange line with our theory model. And fitted equation 8 (green line). Thus finding the coefficients:

Symbol	Quantity	Measured value	Predicted value
α	coefficient	0.070 ± 0.007	0.063
β	coefficient	-0.004 ± 0.020	0.004

TABLE I

VALUES FOR THE COEFFICIENTS OF THE HIGHER POWER TERMS.

Both terms agree with their theoretical counterparts within the ranges of their uncertainty. However, β has a 400% standard error which indicates that the data was not of a high enough range for the higher exponent tuning terms to be properly measured. Or a large change in β did not significantly affect the quality of the fit. This implies that only a quadratic approximation is required for a good prediction below 40 degrees. The periods for larger angles should be measured in future to give the later coefficients more weight in the fit and less uncertainty when finding them.

To further investigate the pendulum's motion a numerical simulation was setup in python using a Runge-Kutta method solving equation 1 for a time period of 5 seconds. A discrete Fourier transform was applied using *numpy.fft.fft*. A clear peak is shown at the fundamental frequency and smaller secondary peak is seen at roughly 3 times the fundamental. Research online regarding why the even harmonic was skipped (2x the fundamental frequency) yielded limited insight but suggested that the symmetry of the motion lead to even harmonics being suppressed. [4]

The *dfft* applied on the actual data produced an almost identical graph 5 to the simulation. The main difference was a slower decay away from the peak and a wider secondary peak at 3.6 Hz. This is consistent with a small amount of random error present in the system, whether that be an error function inherent to the sensor or a small amount of damping due to air resistance.

Perhaps the most interesting element of this analysis was the apparent invariance of the peak frequency with respect to

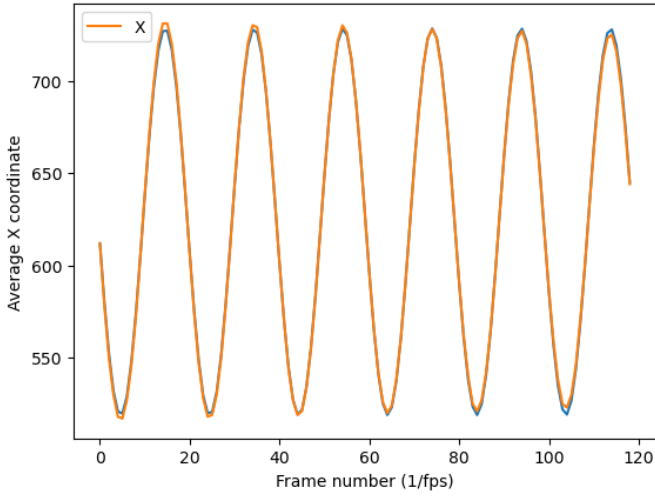


Fig. 3. An example of the x coordinate data from one of the videos and its associated sin curve fit.

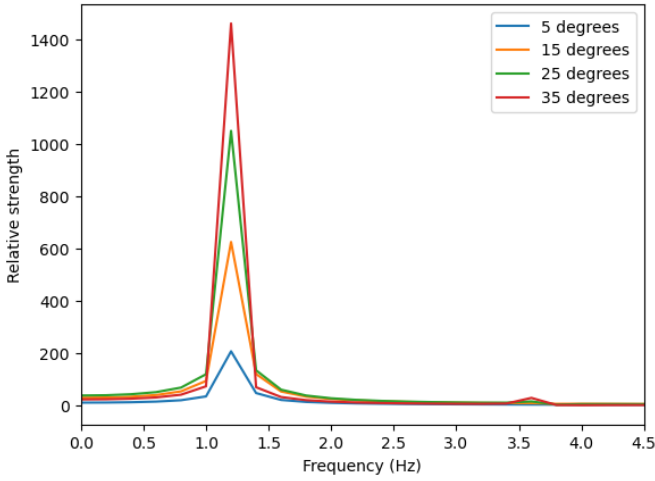


Fig. 4. Theoretical prediction calculated using a numerical simulation in python using a fourth order Runge-Kutta method. It was then discretely Fourier transformed with the `numpy.fft.rfft` function.

amplitude. Whilst the period of oscillation measured by the sin fit increased by $0.02s$ between 5 and 35 degrees. The peak frequency in the Fourier transform shifted negatively by 1 ms .

The fundamental peak frequency of 5 was $(0.823 \pm 0.001)s$ implying a pendulum of length $(0.168 \pm 0.002)m$ which disagrees with the measured length 10 . However it agrees with the length predicted by the theoretical model 11. This provides strong evidence that the shift in time period in 2 between the theoretical and actual values was due to an incorrect measurement of length. This is probably due to neglecting the difference in length between the top of the mass and it's centre of mass. Which the $0.65cm$ difference is approximately equal to.

V. CONCLUSION

This experiment demonstrated that a pendulum's period of oscillation depends on its initial angular displacement. Which limits the use cases for simple harmonic motion

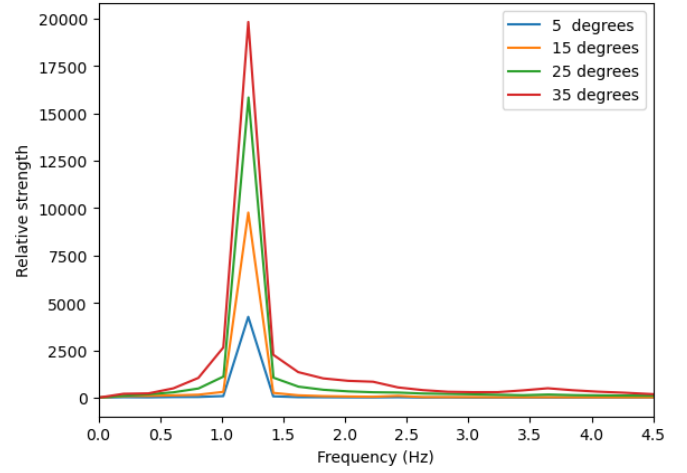


Fig. 5. Discrete Fourier transform of the actual data recorded from the videos.

being assumed, to small angles. It showed that a quadratic approximation was sufficient for angles up to 40 degrees.

ACKNOWLEDGMENT

ChatGPT was used to generate the code for the numerical simulation of the pendulum equation. 1.

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